



Hands on Homework #3 (3D Particle Equilibrium)

EGN 3310 - 0M01 Engineering Analysis: STATICS

MWF, 11:30AM-12:20AM

Submitted by: Group 38

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Hands-On Method (Physical Model)

Before weight applied

For this hand-on, we had everything set up for us, other than the initial heights of the poles. We used the measurements for the NID ending with 8, which makes

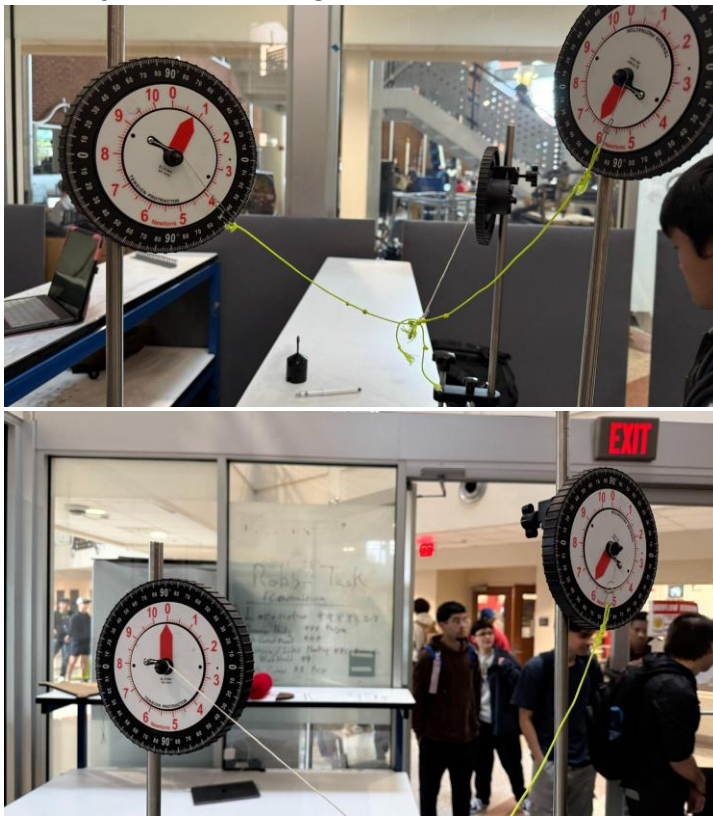
$h_1 = 16\text{in}$

$h_2 = 20\text{in}$

$h_3 = 17\text{in}$

Mass = 500g

We adjusted the heights of the poles, and this was our starting point.



As you might see, the dials didn't start at 0. This is because our stations for h_2 and h_3 were broken, so we had to set something else as the original value.

For h_1 it started at 0N, for h_2 it started at 6.3N and h_3 at 0.8N.

We also had to calculate the location for each pole by setting up a coordinate system on the table. We made the location of the weight as the origin for our convenience.

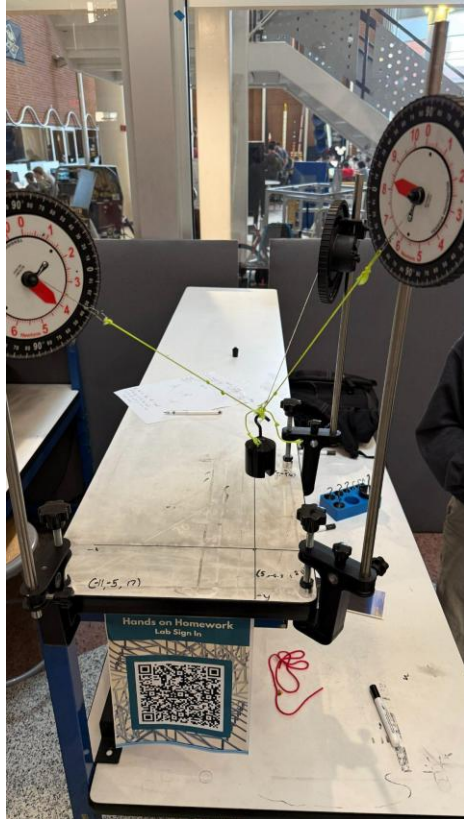
For convenience purpose, we will be naming the poles A,B,C. $A=h_1$, $B=h_2$ and $C=h_3$.

Here are the coordinates of each(in cm)

	X	Y	Z
A	8.89	26.67	40.64
B	12.7	-16.51	50.8
C	-27.94	-12.7	43.18

After weight applied





Now that we applied the weight, we can calculate the other components.

Here are all the other measurements that we took for each pole. This includes the force, angle and length of the strings. Keep in mind that the force measurements we took are based on the fact that we had the forces set as a zero at a larger number due to the wheels being broken.

	Force (N)	Angle	String Length (cm)
A	2.9	34°	30.48
B	2	57°	36.51
C	3.3	35°	35.56

Finally, we measured the sag, and set where we put the weight as the origin. The sag we measured was 22.86 cm, which makes our origin at the point (0,0,22.86).

That concludes the hands-on portion of the homework. We had to get our data in this portion to help us with the following section and so we can compare.

Hand Calculation (Theoretical Method)

1) The weight needed to produce the sag and tensions measured.

Here is the paper with all of our calculations.

1) The weight needed to produce Analytical

A(8.89, 26.67, 40.64) ^{tension}

B(12.7, -16.51, 50.8)

C(-27.94, -12.7, 43.18)

S = (0, 0, 22.86)

Sag

W = <0, 0, -W>

$\sum F_z = 0$

1.55 + 1.60 + 1.82 - W = 0

W = 4.97 N

mass = 0.507 kg

$\vec{SA} = \langle 8.89, 26.67, 17.78 \rangle$

$\vec{SB} = \langle 12.7, -16.51, 27.94 \rangle$

$\vec{SC} = \langle -27.94, -12.7, 20.32 \rangle$

$U_{SA} = \langle \frac{8.89}{33.26}, \frac{26.67}{33.26}, \frac{17.78}{33.26} \rangle$

$U_{SB} = \langle \frac{12.7}{34.85}, \frac{-16.51}{34.85}, \frac{27.94}{34.85} \rangle$

$U_{SC} = \langle \frac{-27.94}{36.81}, \frac{-12.7}{36.81}, \frac{20.32}{36.81} \rangle$

$F_{SA} = F_A \cdot U_{SA} = \langle 0.775, 2.33, 1.55 \rangle N$

$F_{SB} = F_B \cdot U_{SB} = \langle 1.729, -0.947, 1.607 \rangle N$

$F_{SC} = F_C \cdot U_{SC} = \langle -2.50, -1.14, 1.82 \rangle N$

For this portion, we first started by writing all this in cartesian form by getting vectors from the origin to each point, then getting the unit vector and multiplying by the force.

We set S= origin, which includes the sag measurement
In cartesian form, we is what we got

$$F_{sa} = < 0.775, 2.33, 1.55 > N$$

$$F_{sb} = < 0.729, -0.947, 1.60 > N$$

$$F_{sc} = < -2.50, -1.14, 1.82 > N$$

Now since we are finding the weight needed for the tension and sag, we need to set all the forces of the z-components to zero, and solve for weight.

$$\begin{aligned}\Sigma F_z &= 0 \\ 1.55 + 1.60 + 1.82 - W &= 0\end{aligned}$$

Solving for that we get that $W = 4.97 \text{ N/cm}$

Converting this back to grams, we get that the mass is 507g or 0.507kg.

As for percent error,

$$\text{Percent error} = \left| \frac{0.507 - 0.500 \text{ kg}}{0.500 \text{ kg}} \right| \times 100\% = 1.4 \%$$

This is quite low, which means that we did our coordinate system almost perfectly. However, most likely some measurement error caused this small percent error.

2) If the weight and geometry are known, calculate the tension in the cables.

Here is the paper with all our calculations

2) Calculate the tension in the cables

$$W = 0.5 \cdot 9.8 \text{ N}$$

$$W = 4.9 \text{ N}$$

$$\sum F_x = 0 \rightarrow \frac{8.89}{33.26} F_{SA} + \frac{12.7}{34.85} F_{SB} - \frac{27.94}{36.81} F_{SC} = 0$$

$$\sum F_y = 0 \rightarrow \frac{26.67}{33.26} F_{SA} - \frac{16.51}{34.85} F_{SB} - \frac{12.7}{36.81} F_{SC} = 0$$

$$\sum F_z = 0 \rightarrow \frac{17.78}{33.26} F_{SA} + \frac{27.94}{34.85} F_{SB} + \frac{20.32}{36.81} F_{SC} = 4.9 \text{ N}$$

$$F_{SA} = 2.63 \text{ N}$$

$$F_{SB} = 2.80 \text{ N}$$

$$F_{SC} = 2.27 \text{ N}$$

For this portion we know everything and we need to find the tension. Please refer to the previous tables for the geometries, and the weight is 0.500kg.

Using the calculations from the previous question, we can set up a system of equations using each unit vector in order to solve for each tension. $W = 4.9 \text{ N}$.

$$\sum F_x = 0 \rightarrow \frac{8.89}{33.26} F_{SA} + \frac{12.7}{34.85} F_{SB} - \frac{27.94}{36.81} F_{SC} = 0$$

$$\sum F_y = 0 \rightarrow \frac{26.67}{33.26} F_{SA} - \frac{16.51}{34.85} F_{SB} - \frac{12.7}{36.81} F_{SC} = 0$$

$$\sum F_z = 0 \rightarrow \frac{17.78}{33.26} F_{SA} + \frac{27.94}{34.85} F_{SB} + \frac{20.32}{36.81} F_{SC} = 4.9 \text{ N}$$

Solving this using the calculator, we used the method Dr. Zaurin showed in his calculator tip video.

This gave us the following values for the tensions.

$$F_{sa} = 2.63N$$

$$F_{sb} = 2.80N$$

$$F_{sc} = 2.27N$$

Now for each percent error:

$$\text{Percent error } (F_{sa}) = \left| \frac{2.63N - 2.9N}{2.9N} \right| \times 100\% = 9.31\%$$

$$\text{Percent error } (F_{sb}) = \left| \frac{2.80N - 2N}{2N} \right| \times 100\% = 40\%$$

$$\text{Percent error } (F_{sc}) = \left| \frac{2.27N - 3.3N}{3.3N} \right| \times 100\% = 31.2\%$$

These high percent errors most likely stems from the dials being broken that we mentioned at the start. This is most likely true because our unit vector measurements were proven to be right from the percent error in the previous section.

Conclusion/discussion:

Vincent Tremblay (vi474918):

Overall, I think we did a pretty good job finding the values given what we had. This lab really showed how some data can be measured wrong, leading to high percent errors (Tension) and well measured data can lead to low percent error (Weight). As we mentioned, the pulleys were functioning weird, so that led to the high percent error for the calculated tensions. However, working on this was a great way to learn about static equilibrium and setting up systems of equations, further improving our knowledge of statics.

Andrew Tran (an365172):

After finishing our calculations for both parts, I believe that our group produced relatively accurate data given our situation. For calculating the weight of the (500g) mass given the tension forces and cartesian forces, we had a (what we believe to be) low percent error of only 1.4%. However, when calculating the tension forces given the force due to weight and cartesian forces, we had varying percent errors that were significant larger. Regardless, the fact that our results had reasonable amounts of percent error (and thus is somewhat accurate) is evidence of 3d equilibrium and proof of the methods for solving it (systems of equations, setting summation of forces equal to "x" value, etc.).

Kevin Sanchez Ramos (ke987766):

Based on our results, we can see that after all our calculations and measurements, the data comparison from the pulley and our hands-on measurement was not as bad as we expected. Despite the pulley not being in the best shape when we arrived, the percent error of the project was only 1.4% which shows that our math and measurement were almost spot on despite our shortcomings. Overall I believed this experience helped further improve our knowledge in learning about statics as we used previous knowledge to find all the unknowns.